

Task 3: Open Project

Overview

For this open project, you will tackle a more challenging scenario: 20,000 higher-resolution lunar images with even more noise. Start from your code in Task 2 and enhance it to handle this more complex situation. Experiment with different architectures, hyperparameters, loss functions, and training strategies. Your grade will depend on both the effectiveness of your final solution and the thoroughness of your analysis in the report. (See Figure 1 for an example of the task's output.)

Note: Due to the size and complexity of the dataset, you **must** train your models on a GPU on the Euler cluster.

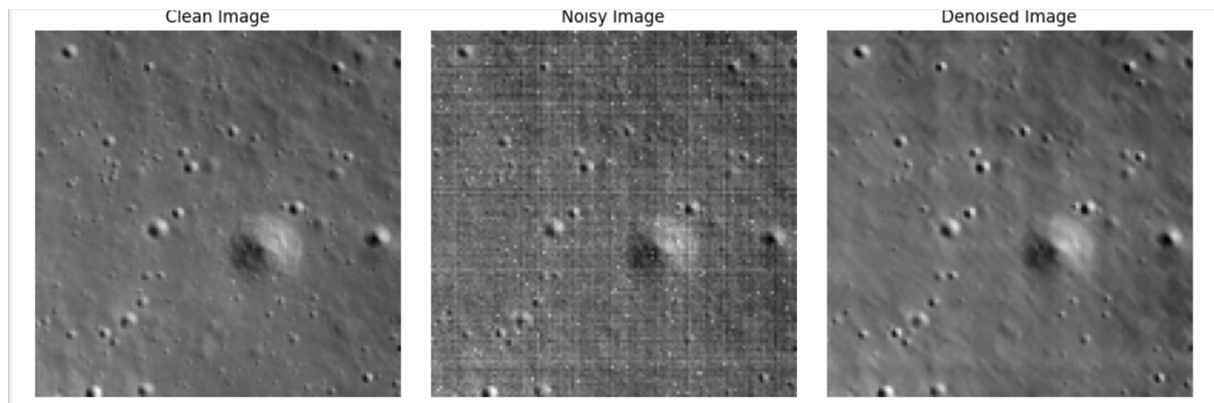


Figure 1) Example output of solution to the problem

Training and Testing Data

- **Training Data:** Provided on Moodle under “Training data”
- **Testing Data:** Noisy images provided only, clean images are held back

Tasks

1. Model Implementation:

- Adapt your code from Task 2 to denoise higher-resolution, noisier data.
- Consider a variety of improvements, including but not limited to:
 - Different optimizers, learning rates, and activation functions.
 - Deeper or more complex architectures (e.g., adding batch normalization, skip connections, ...)
 - Alternate loss functions or regularization techniques.
- Pretrained models are not allowed, but you may draw inspiration from known architectures such as U-Nets or ResNets.
- Ensure your code is GPU-compatible and run training on the Euler cluster.

2. Report Submission (Maximum 3 pages):

Your report should include the following four sections: Introduction, Methods, Results and Discussion. In *yourlastname_yourfirstname_OpenProject_report.pdf*, include:

- A concise description of your final model (its architecture, hyperparameters, optimizer, loss function, ...)

- A discussion of the challenges you faced and how you addressed them.
- Analysis of at least three different configurations you tried (e.g., varying hyperparameters, architectures, or loss functions), explaining what worked better and why.

3. Visualization and Analysis:

- **Cross-Validation:** Perform k-fold cross-validation on your final model and report the training and validation losses for each fold.
- **Configuration Comparisons:** Plot the training and test losses for at least three different configurations.
- **Image Visualizations:** Select two images from the training set and two images from the test set. For each, show the clean, noisy, and denoised images. Comment on the results.

4. Test set evaluation:

- Denoise all the images in the test set and save them as an .npz file

Submission Guidelines

Your submission must include:

1. *yourlastname_yourfirstname_OpenProject.py* (your code)
2. *yourlastname_yourfirstname_OpenProject_report.pdf* (your report)
3. *yourlastname_yourfirstname_prediction.npz* (your prediction on the test set)

Package these into a ZIP file named *yourfirstname_yourlastname_OpenProject_yourleginumber.zip*. Only submissions containing all three components will be graded.

Deadline: 05.06.2026 at 23:59:59